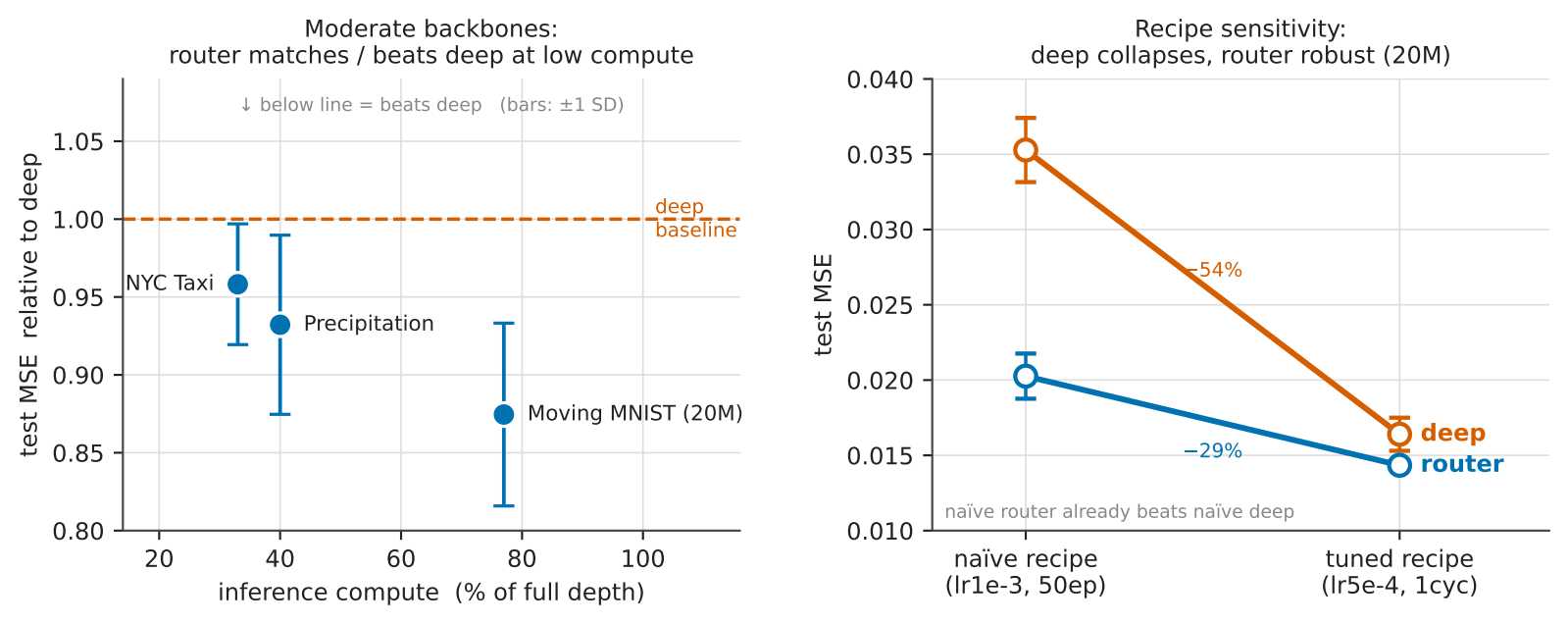
Moderate backbones:
router matches / beats deep at low compute
↓ below line = beats deep (bars: ±1 SD)
test MSE relative to deep
1.05
1.00
0.95
0.90
0.85
0.80
deep
baseline
NYC Taxi
Precipitation
Moving MNIST (20M)
20
40
60
80
100
inference compute (% of full depth)
Recipe sensitivity:
deep collapses, router robust (20M)
test MSE
0.040
0.035
0.030
0.025
0.020
0.015
0.010
−54%
−29%
deep
router
naïve router already beats naïve deep
naïve recipe
(lr1e-3, 50ep)
tuned recipe
(lr5e-4, 1cyc)